

CENTRAL NEBRASKA, NE, USA

WMO: 725520

Lat: 40.961N

Lon: 98.314W

Elev: 1840

StdP: 13.74

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.0	2.0	-10.5	3.3	-1.6	-5.8	4.3	3.3	34.1	23.9	30.6	26.0	10.2	350	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	95.5	74.2	92.4	73.4	89.4	72.3	77.7	89.3	76.2	87.9	74.6	86.0	14.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	137.7	84.5	72.6	129.6	83.0	71.0	122.4	81.3	42.6	89.5	40.9	88.2	39.3	86.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.0	25.9	23.3	-11.5	101.5	3.8	3.4	-14.3	103.9	-16.6	105.9	-18.7	107.8	-21.5	110.3		
			-12.0	81.1	3.8	1.5	-14.8	82.2	-17.0	83.1	-19.1	83.9	-21.8	85.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	25.4	28.8	40.0	50.4	61.3	72.1	76.6	74.0	66.3	52.8	39.5	28.1
	DBStd	20.47	12.17	12.59	11.98	9.91	8.86	7.02	6.00	5.71	8.46	9.53	10.40	11.12
	HDD50	2933	763	596	350	113	12	0	0	3	79	338	679	
	HDD65	6051	1227	1013	776	447	179	19	2	5	85	391	764	1143
	CDD50	3443	0	3	40	124	363	663	823	745	491	167	24	0
	CDD65	1087	0	0	1	9	65	232	360	284	123	13	0	0
	CDH74	11327	0	2	32	164	759	2461	3793	2619	1308	181	8	0
	CDH80	4757	0	0	4	45	267	1035	1766	1089	500	50	1	0
Wind	WSAvg	11.1	11.2	11.8	12.5	13.5	11.9	10.8	9.0	8.7	10.2	11.2	11.2	11.0
Precipitation	PrecAvg	25.7	0.6	0.8	1.6	2.6	4.0	4.0	3.1	2.8	2.6	1.7	1.0	0.8
	PrecMax	39.1	1.7	3.4	6.6	7.3	9.1	14.0	10.7	8.7	9.0	6.0	3.8	3.5
	PrecMin	11.5	0.0	0.1	0.0	0.1	0.4	0.0	0.2	0.6	0.2	0.0	0.0	0.0
	PrecStd	6.2	0.4	0.6	1.4	1.5	2.2	2.7	2.0	1.7	2.0	1.4	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.0	68.5	78.3	85.9	92.2	97.3	100.4	98.3	94.4	86.6	74.3	60.1
		MCWB	45.9	50.2	56.1	62.0	68.1	72.7	75.0	73.5	70.0	65.0	55.0	46.0
	2%	DB	52.3	58.7	71.6	78.5	86.7	93.1	96.2	93.5	89.9	78.7	67.3	53.2
		MCWB	41.3	45.7	53.7	59.7	67.0	72.7	75.9	74.4	69.7	60.6	51.4	43.3
	5%	DB	46.7	52.4	65.0	73.2	82.3	89.6	92.7	89.7	86.0	73.9	61.1	48.3
		MCWB	38.5	43.0	51.8	56.8	64.8	71.5	75.1	74.1	68.5	58.5	49.4	40.1
	10%	DB	41.4	46.4	59.2	67.8	77.6	86.1	89.3	86.1	81.7	69.2	55.6	43.2
		MCWB	35.8	39.5	48.6	54.5	63.0	69.9	74.2	72.6	66.2	56.6	46.3	37.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.3	52.2	61.6	65.3	72.4	77.6	80.4	78.8	74.4	68.1	58.8	49.6
		MCDB	60.0	63.4	69.4	78.7	85.1	89.0	91.7	89.7	87.1	79.6	66.3	57.7
	2%	WB	42.1	47.0	57.4	61.7	69.6	75.3	78.4	77.1	72.3	64.5	54.2	43.8
		MCDB	51.1	57.6	66.6	73.4	81.6	88.3	90.2	88.2	84.4	73.0	63.9	52.2
	5%	WB	39.0	43.2	53.0	58.8	67.3	73.5	77.0	75.3	70.6	61.1	50.4	40.5
		MCDB	45.8	52.2	63.6	71.1	78.8	85.9	88.9	86.4	82.1	70.3	60.2	47.7
	10%	WB	36.1	39.5	48.9	55.7	64.6	71.8	75.3	73.7	68.6	58.2	46.5	37.1
		MCDB	41.2	46.2	59.2	66.3	75.0	83.3	87.0	84.1	78.7	66.8	55.3	43.0
Mean Daily Temperature Range	5% DB	MDBR	19.7	20.1	23.0	23.3	21.8	22.1	21.3	21.0	24.0	23.7	22.3	19.0
		MCDBR	27.6	29.1	33.0	32.3	28.5	26.3	25.3	25.3	27.6	30.7	31.1	26.4
		MCWBR	18.2	18.3	18.3	16.8	12.7	10.8	10.8	10.8	11.6	15.2	17.9	17.6
	5% WB	MCDBR	26.2	28.3	28.9	28.4	25.0	23.2	22.6	22.4	23.8	24.5	28.8	25.5
		MCWBR	17.8	18.3	17.0	16.5	12.6	11.2	11.5	10.9	11.4	14.1	17.5	17.8
Clear-Sky Solar Irradiance	taub	0.274	0.280	0.307	0.345	0.370	0.385	0.400	0.397	0.362	0.320	0.281	0.271	
	taud	2.483	2.494	2.452	2.364	2.368	2.375	2.349	2.348	2.404	2.504	2.524	2.495	
	Ebn at Noon	285	299	301	294	287	282	277	274	278	280	280	277	
	Edh at Noon	23	27	32	37	38	38	39	38	33	27	23	22	
All-Sky Solar Radiation	RadAvg	694	964	1293	1635	1899	2174	2137	1854	1506	1077	768	588	
	RadStd	63	62	111	90	121	137	101	110	95	108	54	58	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page